

Name: [REDACTED]

Test- Kinematics TI

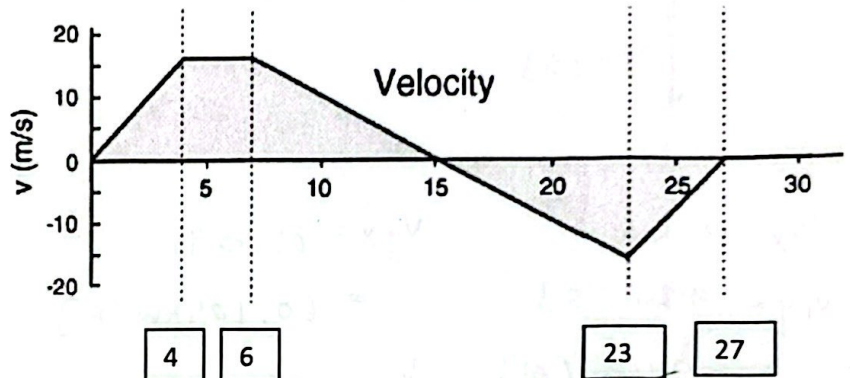
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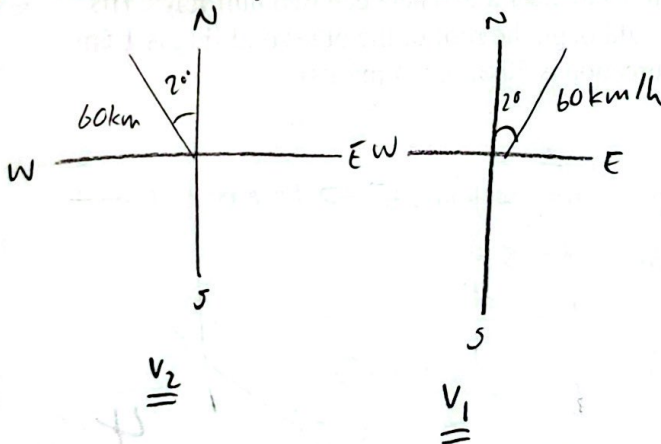
Clearly show ALL YOUR WORK for full credit. Isolate for the unknown variable as required, include units and directions. Round your final answer to 3 s.f.

1. A velocity time graph depicts the motion for an object; take positive direction to be **NORTH** and time to **SECONDS**.



- a. State the time(s) when the object is speeding up in the negative direction. 15s - 23s (1 mark)
- b. State the time(s) when the object is slowing down in the positive direction 10s - 15s (1 mark)
- c. Describe the motion from 4 to 6 seconds. the object is moving at a constant velocity of 15 m/s [N] (1 mark)

2. Over a 15 minute period a car travels on a winding road. The car's initial velocity is 60 km/h [N 20° E]. The final velocity is 60 km/h [N 20° W]. Calculate the average acceleration of the car. (4 marks)



$$a_{av} = \frac{v_2 - v_1}{t}$$

$$v_{Tx}: v_2 - v_1 = 20.521 \text{ km/h [W]} - (-20.521 \text{ km/h [W]}) = 41.042 \text{ km/h [W]}$$

$$v_{2x}: 60 \sin 20$$

$$= 20.521 \text{ km/h [W]}$$

$$v_{1x}: 60 \sin 20$$

$$= 20.52 \text{ km/h [E]}$$

$$= -20.521 \text{ km/h [W]}$$

$$v_{Ty}: v_2 - v_1$$

$$= 56.382 \text{ km [W]} - 56.382$$

$$\text{km/h [W]}$$

$$= 0 \text{ km/h [W]}$$

$$v_{2y}: 60 \cos 20$$

$$= 56.382 \text{ km/h [N]}$$

$$v_{1y}: 60 \cos 20$$

$$= 56.382 \text{ km/h [N]}$$

$$v_T = \sqrt{41.042^2 + 0^2}$$

$$= 41.042 \text{ km/h [W]}$$

$$0.25 \text{ h}$$

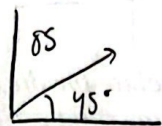
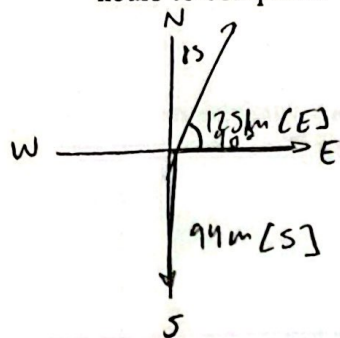
$$\vec{a}_{av}$$

$$= 164.17 \text{ km/h}^2$$

$$\tan^{-1} \left(\frac{0}{41.042} \right) = 0$$

$$a_{av} = 164.17 \text{ km/h}^2 \text{ [W]}$$

3. A cruise ship sails 94.0 km [S], then 125 km [E] and finally 85.0 km [E 45° N]. The trip takes 6.75 hours to complete. Calculate the average velocity of the sailboat. (4 marks)



$$\begin{aligned} v_{1x} &= 0 \text{ km} \\ v_{1y} &= 94 \text{ km [S]} \\ &= -94 \text{ km [N]} \\ v_{2x} &= 125 \text{ km [E]} \\ v_{2y} &= 0 \text{ km} \end{aligned}$$

$$\begin{aligned} v_{3x} &= 85 \cos 45 \\ &= 60.104 \text{ km [E]} \\ v_{3y} &= 85 \sin 45 \\ &= 60.104 \text{ km [N]} \end{aligned}$$

$$\begin{aligned} v_{Tx} &= 0 + 60.104 \text{ km [E]} + 125 \text{ km [E]} \\ &= 185.10 \text{ km [E]} \\ v_{Ty} &= 0 + (-94 \text{ km [N]}) + 60.104 \text{ km [N]} \\ &= -33.896 \text{ km [N]} \end{aligned}$$

$$v_T = \sqrt{(33.896 \text{ km [S]})^2 + (185.10 \text{ km [E]})^2}$$

$$= 188.18 \text{ km}$$

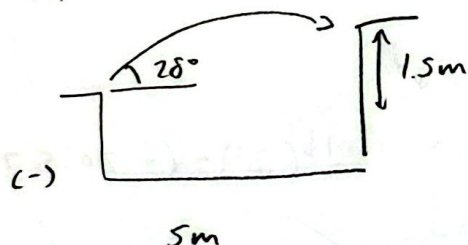
$$\tan^{-1} \frac{33.896}{185.10} = \theta$$

$$\theta = 10^\circ \text{ [E } 10^\circ \text{ S]}$$

$$v = \frac{188.18 \text{ km}}{6.75 \text{ h}} = 27.87$$

$$v = 27.9 \text{ km/h [E } 10^\circ \text{ S]}$$

4. Christian continues his daredevil adventures and attempts to clear a gap between two buildings. His launch angle is 28 degrees, and the building is 5m away, although the roof of the other building is 1.5m higher. Does he make it if his initial velocity in the x direction is 50 km/h? (4 marks)



$$\begin{aligned} v_{ix} &= v_i \sin 28 \\ &= 15.730 \text{ (sin } 28) \\ &= 7.3897 \text{ m/s [↑]} \end{aligned}$$

$$\begin{aligned} v &= 50 \text{ km/h} \Rightarrow 13.8 \text{ m/s [→]} \\ d &= 5 \text{ m} \\ t &= \frac{d}{v} \\ t &= \frac{5 \text{ m [→]}}{13.8 \text{ m/s [→]}} \end{aligned}$$

$$v_{ix} = 50 \text{ km/h} \Rightarrow 13.8 \text{ m/s [→]}$$

$$\frac{v_i \cos 28}{\cos 28} = 13.8 \text{ m/s}$$

$$v_i = 15.730 \text{ m/s [→ } 28^\circ \text{ ↑]}$$

$$a = -9.81 \text{ m/s}^2 \text{ [↑]}$$

$$d = 1.5 \text{ m [↑]}$$

$$t = 0.3623 \text{ s}$$

$$d = ?$$

$$\Delta d = v_i \Delta t + \frac{1}{2} a \Delta t^2$$

$$\Delta d = (7.3897 \text{ m/s [↑]})(0.3623 \text{ s}) + \frac{1}{2} (-9.81 \text{ m/s}^2 \text{ [↑]})$$

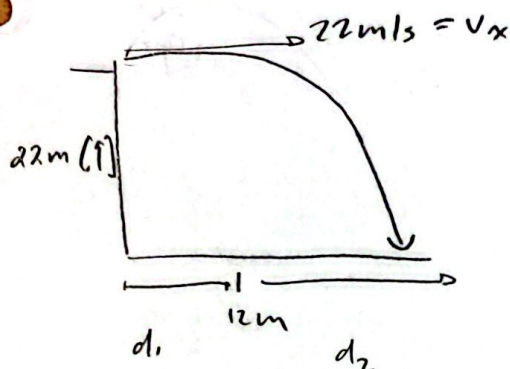
$$\Delta d = 2.03 \text{ m}$$

∴ he does make the jump as he jumps 2.03 m [↑]

* can assume x is true then solve (0.3623 s) for sdy or vice versa

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A quarterback throws a football from the edge of a 22.0 m high cliff with a constant horizontal velocity of 22 m/s. As the ball is thrown, a player at the bottom of the cliff 12 m away from its base, accelerates at 3.5 m/s² to try and catch the ball. Show by calculation if it is possible for the player to catch the ball. The player has a top speed of 9 m/s. (5 marks)



$$y$$

$$a = -9.81 \text{ m/s}^2 (\uparrow)$$

$$d = -22 \text{ m} (\uparrow)$$

$$v_i = 0 \text{ m/s}$$

$$\Delta d = v_i \Delta t + \frac{1}{2} a t^2$$

$$t = \sqrt{\frac{\Delta d}{\frac{1}{2} a}}$$

$$= \sqrt{\frac{-22 \text{ m} (\uparrow)}{\frac{1}{2} (-9.81 \text{ m/s}^2 (\uparrow))}}$$

$$= 2.1178 \text{ s} = 2.12 \text{ s}$$

$\therefore$ the runner cannot catch the ball because he runs a total of 19.8 m ($\rightarrow$) while the ball travels 46.6 m ($\rightarrow$) in 2.12 s

$$x$$

$$t = 2.1178 \text{ s}$$

$$v = 22.0 \text{ m/s} (\rightarrow)$$

$$d = vt$$

$$d = (22.0 \text{ m/s} (\rightarrow)) (2.1178 \text{ s})$$

$$= 46.592 \text{ m} (\rightarrow) \leftarrow \text{horiz. displacement}$$

$$= 46.6 \text{ m} (\rightarrow)$$

player

$$v_i = 0 \text{ m/s} (\rightarrow)$$

$$\text{top speed } (v_2 \text{ m/s}) = 9 \text{ m/s}$$

$$a = 3.5 \text{ m/s}^2 (\rightarrow)$$

$$t = 2.1178 \text{ s}$$

$$d = ? (+12 \text{ m}) \leftarrow d_1$$

$$\Delta d = v_i \Delta t + \frac{1}{2} a t^2$$

$$= \frac{1}{2} (3.5 \text{ m/s}^2 (\rightarrow)) (2.1178)^2$$

$$= 7.8988 \text{ m} (\rightarrow)$$

$$d_2 + d_1 = d_T$$

$$d_T = 7.8988 \text{ m} (\rightarrow) + 12 \text{ m} (\rightarrow)$$

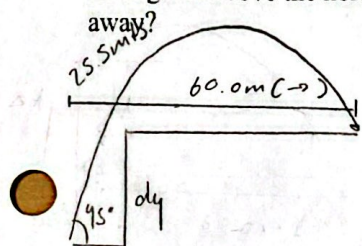
$$= 19.8988 \text{ m} (\rightarrow) = 19.8 \text{ m} (\rightarrow)$$

$\therefore$ horiz. displacement of runner

Name

the magnitude of $v_i > v_f$. v_i is 25.5 m/s ($\rightarrow 45^\circ \uparrow$), and v_f is 23.2 m/s ($\rightarrow 39^\circ \downarrow$). This is because the projectile has a positive y -displacement (lands higher than it was launched), so there is less time for gravity ($-9.81 \text{ m/s}^2 = \text{acceleration}$) to increase the v_f value.

1. Assuming the initial velocity of a Nerf Projectile released during our projectile lab was 25.5 m/s at an angle of 45 degrees above the horizon, how high would the arrow have to be launched from the ground to land 60.0 m away?



$$\begin{aligned} x & \\ v &= 25.5 \cos 45 \\ &= 18.031 \text{ m/s } (\rightarrow) \\ d &= 60 \text{ m } (\rightarrow) \\ t &= \frac{d}{v} \\ &= \frac{60 \text{ m } (\rightarrow)}{18.031 \text{ m/s } (\rightarrow)} \\ &= 3.3276 \text{ s} \end{aligned}$$

$$\begin{aligned} y & \\ t &= 3.3276 \text{ s} \\ v_i &= 25.5 \sin 45 \\ &= 18.031 \text{ m/s } (\uparrow) \\ a &= -9.81 \text{ m/s}^2 (\uparrow) \\ d &= ? \\ \Delta d &= v_i t + \frac{1}{2} a t^2 \\ &= (18.031 \text{ m/s } (\uparrow))(3.3276) \\ &\quad + \frac{1}{2}(-9.81 \text{ m/s}^2 (\uparrow))(3.3276)^2 \\ &= 5.7926 \text{ m } (\uparrow) \\ &= 5.79 \text{ m } // \end{aligned}$$

the projectile would have to be launched at a height of 5.79 m from below the ground.

b. Calculate the final velocity of the Nerf Projectile just before it hits the ground. Briefly explain the difference compared to the initial velocity.

$$\begin{aligned} x & \\ v &= 25.5 \cos 45 \\ &= 18.031 \text{ m/s } (\rightarrow) \end{aligned}$$

$$\begin{aligned} y & \\ a &= -9.81 \text{ m/s}^2 (\uparrow) \\ t &= 3.3276 \text{ s} \\ v_i &= 18.031 \text{ m/s } (\uparrow) \\ v_f &= v_i + at \end{aligned}$$

$$\begin{aligned} v_f &= \sqrt{19.613^2 + 18.031^2} \\ &= 23.208 \text{ m/s} \\ \tan^{-1} \left(\frac{19.613}{18.031} \right) &= \theta \\ \theta &= 39^\circ (\rightarrow 39^\circ \downarrow) \end{aligned}$$

the final velocity of the projectile is 23.2 m/s ($\rightarrow 39^\circ \downarrow$)

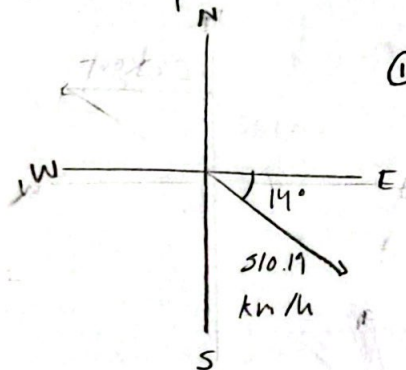
$$\begin{aligned} &= 18.031 \text{ m/s } (\uparrow) + (-9.81 \text{ m/s}^2 (\uparrow))(3.3276 \text{ s}) \\ &= -19.613 \text{ m/s } (\uparrow) \Rightarrow 19.613 \text{ m/s } (\downarrow) \end{aligned}$$

2. An airplane takes off from London, England at 11:05 am and is scheduled to arrive at Prague, Czech Republic at 12:30 pm. The pilot is alerted to a strong wind of 65 km/h [N 30°W].

London is 180 km North and 700 km west of Prague. If the wind velocity does not change the entire trip, determine the airspeed and heading that the pilot must maintain in order to reach the destination on time.



P = plane E = earth A = air



$$\textcircled{1} d_{PE} = \sqrt{180^2 + 700^2} = 722.77 \text{ km}$$

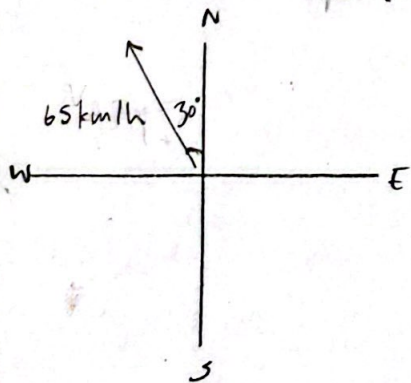
$$\theta = \tan^{-1}\left(\frac{180}{700}\right) = 14.421^\circ \rightarrow [E 14.5^\circ]$$

$$\text{time: } 12:30 \text{ pm} - 11:05 \text{ am} = 85 \text{ min} \Rightarrow 1.4167 \text{ h}$$

$$\textcircled{2} v_{PEy} = (510.19 \text{ km/h})(\sin 14.4) = 126.87 \text{ km/h [S]} \quad d_{PE} = 722.77 \text{ km [E } 14.5^\circ]$$

$$PE = \frac{722.77 \text{ km [E } 14.5^\circ]}{1.4167 \text{ h}}$$

$$v_{PEx} = (510.19 \text{ km/h})(\cos 14.4) = 494.16 \text{ km/h [E]} \quad = 510.19 \text{ km/h [E } 14.5^\circ]$$



$$\textcircled{3} v_{PE} = v_{PA} + v_{AE}$$

$$v_{PA} = v_{PE} - v_{AE}$$

$$v_{PAy} = (126.87 \text{ km/h [S]}) - (-56.292 \text{ km/h [S]}) = 183.162 \text{ km/h [S]}$$

$$v_{PAx} = (494.16 \text{ km/h [E]}) - (-32.5 \text{ km/h [E]}) = 526.66 \text{ km/h [E]}$$

$$v_{PAF} = \sqrt{526.66^2 + 183.16^2} = 557.600 \text{ km/h}$$

$$\tan^{-1}\left(\frac{183.16}{526.66}\right) = \theta = [E 19.5^\circ]$$

the airspeed the pilot must maintain is 558 km/h [E 19°S]